Om Pathak

Bachelor of Engineering (2021-25)
Artificial intelligence and Machine learning
BNM Institute of Technology, Bengaluru

+91-8334061827
ompathak761@gmail.com
github.com/pathakom09
www.linkedin.com/in/om-pathak-72b3b8229/

Education

BNM Institute of Technology *2021-2025*
Artificial intelligence and Machine learning CGPA: **8.56**

Notable Courseworks: NLP, BIG Data, Machine Learning, Data Structures and Algorithms, Operating System, AI, Cloud Computing, Database and Management System, Computer Networks

Projects

Breathalyzer - Using respiratory metrics *2024*

- **Developed** a ML model for respiratory metrics using **Machine Learning algorithm, Python and StreamLit** and Implemented **Random Forest algorithm** to classify **1.1M respiratory data** records with **95% accuracy**.
- **Ensured data reliability** and consistency for **780K** and **308K** records through advanced **preprocessing** techniques. and **Enhanced** data processing efficiency by **30%**, and reduced runtime by **40%** using **Pandas** and **NumPy**.

Diabetic Retinopathy Prediction Model (GAN + CNN + GPT-4o) *2025*

- **Developed** a medical image analysis model for detecting **diabetic retinopathy** using GANs, CNNs, and Integrated LLM Model - GPT-4 for faster report generation with **97%** accurate prediction results.
- **Developed** Frontend UI for patient details upload and integrated backend APIs to upload the data on **MongoDB, Cloudinary and AWS S3 bucket** for analysis and prediction.

Movie Recommendation System *2023*

- **Built** a Movie recommendation system using **Python, Streamlit, NLP, Moviedataset, APIAchievements** and Merged and **processed** over **10,000 records** from two datasets, reducing **missing values** by **80%**.
- **Key features** like **genres, keywords, cast, and crew** are transformed into a **5,000-word** feature **vector space**using the **"bag of words"** model and implemented **cosine similarity** on that **5,000-dimensional vector space**,
- **The system achieves** over **90% accuracy** in recommending similar movies.

Expense Tracker System *2024*

- **Cloud Based** Expense Tracker Web Application using **REACT, Firebase, Fullstack** and Constructed the frontend of the website using **React. Designed UI using HTML, SCSS and JavaScript**
- **Integrated Firebase** as the cloud platform for **real-time data synchronization** and **secure storage**

Internship Experience

Data Science (Analyst) at S.L.N Engineers *Nov 2024, 3 months*
Data Science (Analyst) Intern **Bengaluru**

- **Cleaned** and **structured** IT records using Python and SQL by removing duplicates, standardizing formats, and organizing data into relational databases..
- **Optimized** relational databases, improving query performance through indexing, joins, and normalization techniques, and **automated** data integration enabling real-time insights and website synchronization.

Technical Skills

Languages: Java, Python, C, SQL, JavaScript, HTML, CSS
Cloud/Databases: MySQL, Azure OpenAI Service, AWS(basic), GITHub, PHPmyAdmin
Libraries: Pandas, Numpy, PyTorch, Keras, Seaborn, Scikit-learn, OpenCV, Keras, Tensorflow
Developer Tools: Cursor AI, Visual Studio, Jupyter, VS Code, NetBins
Data Visualization: Power-BI, Matplotlib
Knowledge of DevOps, LLM, AI Agents, APIs, React, Node and Express.

Certifications

Udemy Online Certification in **AI and Cognitive Science** *2025*
Infosys Spring Board Certification in Introduction to **Data Science** and Exploratory **Data Analysis** *2022*
Linkedin Learning Certification of **Azure AI for Developers: Building AI Agents** *2025*

Achievements(co-curricular and extra-curricular)

IBM Hackathon Selected as top candidate from over **750 participants** for Exemplary performance. *2024*

IIT Kharagpur Hackathon Achieved **pre-finalist** status in a competitive **Data science hackathon.** *2024*

Clubs Q-quotient (quizzing club), BNMIT MUN at BNMIT